

Suppose when $n = k$, for each $\{\alpha_i\}_{i=1}^k \in \mathbb{R}_+^k$ such that $\sum_{i=1}^k \alpha_i = 1$, we have

$$\delta_{z_1} \succsim \alpha_1 \delta_{z_1} + \cdots + \alpha_k \delta_{z_k} \succsim \delta_{z_k}. \quad (1)$$

W.w.t.p., when $n = k + 1$, for each $\{\beta_i\}_{i=1}^{k+1} \in \mathbb{R}_+^{k+1}$ such that $\sum_{i=1}^{k+1} \beta_i = 1$, we have

$$\delta_{z_1} \succsim \beta_1 \delta_{z_1} + \cdots + \beta_k \delta_{z_k} + \beta_{k+1} \delta_{z_{k+1}} \succsim \delta_{z_{k+1}}. \quad (2)$$

We will only prove the second binary relation of (2) (the proof for the first one is similar), *i.e.*, for each $\{\beta_i\}_{i=1}^{k+1} \in \mathbb{R}_+^{k+1}$ such that $\sum_{i=1}^{k+1} \beta_i = 1$, we have

$$\beta_1 \delta_{z_1} + \cdots + \beta_k \delta_{z_k} + \beta_{k+1} \delta_{z_{k+1}} \succsim \delta_{z_{k+1}}. \quad (3)$$

Case 1: when $\beta_{k+1} = 0$, from the second binary relation of (2), we know that

$$\beta_1 \delta_{z_1} + \cdots + \beta_k \delta_{z_k} \succsim \delta_{z_k}. \quad (4)$$

We also have

$$\delta_{z_k} \succsim \delta_{z_{k+1}}. \quad (5)$$

By transitivity (from A1), (4) and (5) implies

$$\beta_1 \delta_{z_1} + \cdots + \beta_k \delta_{z_k} \succsim \delta_{z_{k+1}}. \quad (6)$$

Considering that $\beta_{k+1} = 0$, (6) is the same as (3).

Case 2: when $\beta_{k+1} = 1$, (3) also holds since completeness (A1) implies $\delta_{z_{k+1}} \succsim \delta_{z_{k+1}}$.

Case 3: when $\beta_{k+1} \in (0, 1)$,

$$\sum_{i=1}^{k+1} \beta_i \delta_{z_i} = (1 - \beta_{k+1}) \left(\frac{\beta_1}{1 - \beta_{k+1}} \delta_{z_1} + \cdots + \frac{\beta_k}{1 - \beta_{k+1}} \delta_{z_k} \right) + \beta_{k+1} \delta_{z_{k+1}}. \quad (7)$$

Let

$$\pi \equiv \frac{\beta_1}{1 - \beta_{k+1}} \delta_{z_1} + \cdots + \frac{\beta_k}{1 - \beta_{k+1}} \delta_{z_k}. \quad (8)$$

π is a lottery since for each $i = 1, \dots, n$, $\frac{\beta_i}{1 - \beta_{k+1}} \geq 0$, and $\sum_{i=1}^k \frac{\beta_i}{1 - \beta_{k+1}} = 1$.

By the second binary relation of (2) we know that $\pi \succsim \delta_{z_k}$. Since $\delta_{z_k} \succsim \delta_{z_{k+1}}$, we have

$$\pi \equiv \frac{\beta_1}{1 - \beta_{k+1}} \delta_{z_1} + \cdots + \frac{\beta_k}{1 - \beta_{k+1}} \delta_{z_k} \succsim \delta_{z_{k+1}}. \quad (9)$$

Since $1 - \beta_{k+1} \in (0, 1)$, by IA (A3), (9) implies that

$$(1 - \beta_{k+1})\pi + \beta_{k+1} \delta_{z_{k+1}} \succsim \delta_{z_{k+1}}. \quad (10)$$

Therefore, (3) also holds when $\beta_{k+1} \in (0, 1)$.

iii. Let $\bar{\pi} = \delta_{z_1}$, $\underline{\pi} = \delta_{z_{k+1}}$.

For the preference: for all $\pi, \pi' \in \Delta(Z)$, $\pi \succsim \pi'$ if and only if $\pi = \pi'$; when $\pi \neq \pi'$, we cannot have $\pi \succsim \pi'$ or $\pi' \succsim \pi$. We claim that this preference is continuous.

Proof: W.w.t.p. that for all $\pi, \pi', \pi'' \in \Delta(Z)$, the sets

$$C \equiv \{a \in [0, 1] : a\pi + (1-a)\pi' \succsim \pi''\}$$

$$C' \equiv \{a \in [0, 1] : a\pi'' \succsim \pi + (1-a)\pi'\}$$

are closed sets. If there some number $a \in [0, 1]$ such that $a\pi + (1-a)\pi' \succsim \pi''$ and $\pi'' \succsim \pi + (1-a)\pi'$, then $\pi'' = a\pi + (1-a)\pi'$, i.e.,

$$(a\pi_1 + (1-a)\pi'_1, a\pi_2 + (1-a)\pi'_2, a\pi_3 + (1-a)\pi'_3) = (\pi''_1, \pi''_2, \pi''_3).$$

which is equivalent to ¹

$$a\pi_1 + (1-a)\pi'_1 = \pi''_1 \iff (\pi_1 - \pi'_1)a = \pi''_1 - \pi'_1, \quad (i)$$

and

$$a\pi_2 + (1-a)\pi'_2 = \pi''_2 \iff (\pi_2 - \pi'_2)a = \pi''_2 - \pi'_2, \quad (ii)$$

The system of equations (i) and (ii) may have no solution, or just one solution, or infinite many solutions (any number could be a solution when $\pi = \pi' = \pi''$). So the sets C and C' should be an empty set, or a singleton set, or $[0, 1]$. They are both closed sets.

¹when (i) and (ii) hold, then $a\pi_3 + (1-a)\pi'_3 = \pi''_3$ since the summation of probabilities of a lottery should be 1.